

Inventory and Shop System Proposal

1 Proposal

For this project, I want to create an inventory and shop system, which is the first thing that comes into my mind when I think about gameplay loops and game progression. The inventory part of the system will allow players to pick up items around the game's world space, and the inventory UI will be an $n \times n$ sliced square, the size of which is upgradable. The items will initially be shown as the same dots until players pick them up, after which they'll have to drag the item sprites into the inventory menu, and sort their placements so that they can fit in. The shop system will have its own interface with a trader menu and player menu, players can sell their items in exchange for currency, then use it to buy equipment or upgrades from the shop, there will also be a sell all option available to clear up the inventory.

The inventory and shop will mainly consist of scripts that operate independently from other game systems. Scripts for the inventory should work if they're connected to a player object, either through references from another script or being attached directly. Scripts for the shop will only require inventory scripts for player menu and item exchanges, thus ensuring modularity. All files of the system will be included in a single folder separated from the rest, so that they won't risk overwriting others' works. This should guarantee interoperability as the system is self-contained and will not alter the functions of others. For customizability, there will be variables in scripts that control items' prices, quantities and spawn rates which can all be changed easily, the shop system will use arrays to insert entries into the trade menu, so others can add new items or upgrades following the same structure. It'll also allow altering objects representing the shop, such as the sprite for the shopkeeper.

For the demonstration, I'll set up a 2D scene including grid and foreground and background tilemaps, two objects representing the player and the shop, and randomly spawned items placed in different locations. The player will have health, movement speed and jump height, and can go around and pick up items using the inventory, then go back to the shop for selling and buying, the shopkeeper will have upgrades that can alter the player's variables, which should make visible difference once applied.

For plan B should the inventory development not progress well, I will abandon the design on fitting the items into an $n \times n$ inventory. Instead, I will make the inventory capacity an integer to determine how much items the player can carry, and when players pick up an item, they won't need to drag it into a separate inventory UI, as the item sprite will now automatically appear on the interface at the bottom right corner, which showcases items in a straight line.

Backup projects:

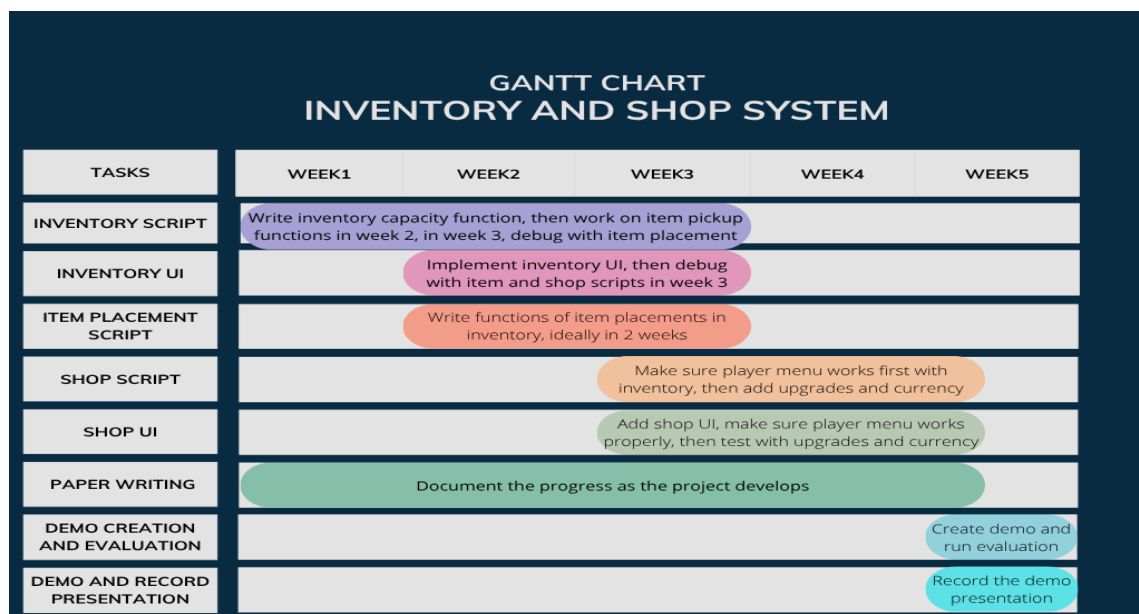
1. Procedural grass generator, able to generate grass with different sizes and colors and corresponding geometry, then automatically place them in the game world with adjustable positions.
2. Cloud rendering system, rendering a variety of clouds with different shapes and sizes, while also capable of light reflection so that their colors alternate naturally depending on the lighting.

2 Related Work

An example of this system would be the inventory in the 2023 game Dredge[1]. In Dredge, players can find fishing spots on the ocean, and after playing a mini game, obtain items with different shapes and sizes, then arrange placements inside inventory to fit the capacity. One thing I'd love to take from it is being able to see shapes of items before picking them up, though I can see difficulties such as generating shapes for all items or ways to trigger the function. An example like my plan B would be The Binding of Isaac: Rebirth[2], where there is no inventory menu, instead items players collected can be listed on screen through its 'Found HUD' option. I'd like to add a similar mechanic to allow the use of items directly with input keys, but how the shop menu will react to this usage may be a problem.

3 Plan

The only external resource the project requires should be a tilemap set containing sprites for the scene, the player and the shop objects, as well as items and the icons for upgrades.



Potential difficulties:

- Since items are polyominoes in this inventory system. I need to learn to write a data model that supports multiple rotations and a consistent pivot.
- It'll be difficult to design the shop system to work with this inventory, I need to figure out how to drag-and-drop between panels, avoiding issues such as duplication and ghosting.

4 References

[1] *Dredge*, Black Salt Games; published by Team17, Windows PC, 2023. [Online]. Available: <https://store.steampowered.com/app/1562430/DREDGE/>

[2] *The Binding of Isaac: Rebirth*, Nicalis, Windows PC, 2014. [Online]. Available: https://store.steampowered.com/app/250900/The_Binding_of_Isaac_Rebirth/